

## **Appendix B. Questionnaire Used in Experiment 2**

### **Experiment 2-1: Grip**

#### **1) Wooden Box**

1. I felt as if I was grasping a real object.  
Strongly disagree — Strongly agree
2. The tactile sensation of the object was similar to that of a real object.  
Strongly disagree — Strongly agree
3. Grasping the object with my hand felt natural.  
Strongly disagree — Strongly agree
4. How hard did the grasped object feel?  
Very soft — Very hard
5. How well did the perceived hardness match the appearance of the virtual object?  
Did not match at all — Matched very well
6. How intense was the overall thermal sensation (warmth/heat)?  
Not felt at all — Felt very strongly

#### **2) Coffee Mug**

1. I felt as if I was grasping a real object.  
Strongly disagree — Strongly agree
2. The tactile sensation of the object was similar to that of a real object.  
Strongly disagree — Strongly agree
3. Grasping the object with my hand felt natural.  
Strongly disagree — Strongly agree
4. How hard did the grasped object feel?  
Very soft — Very hard
5. How well did the perceived hardness match the appearance of the virtual object?  
Did not match at all — Matched very well
6. How intense was the overall thermal sensation (warmth/heat)?  
Not felt at all — Felt very strongly

## Experiment 2-2: Spread

### 1) Sand

1. The resistance I felt while opening my hand felt realistic.  
Strongly disagree — Strongly agree
2. The pressure felt on my fingers was similar to that in the real world.  
Strongly disagree — Strongly agree
3. Opening my hand in the sand felt natural.  
Strongly disagree — Strongly agree
4. How strong did the resistance/viscosity of the sand feel while opening your hand?  
Not strong at all — Very strong
5. How well did the perceived viscosity match the virtual medium (sand)?  
Did not match at all — Matched very well
6. How intense was the overall thermal sensation (warmth/heat)?  
Not felt at all — Felt very strongly

### 2) Water

1. The resistance I felt while opening my hand felt realistic.  
Strongly disagree — Strongly agree
2. The pressure felt on my fingers was similar to that in the real world.  
Strongly disagree — Strongly agree
3. Opening my hand in the sand felt natural.  
Strongly disagree — Strongly agree
4. How strong did the resistance/viscosity of the sand feel while opening your hand?  
Not strong at all — Very strong
5. How well did the perceived viscosity match the virtual medium (sand)?  
Did not match at all — Matched very well
6. How intense was the overall thermal sensation (warmth/heat)?  
Not felt at all — Felt very strongly

### **Presence Questionnaire (Adapted from the IPQ)**

1. I had a sense of being there in the virtual environment.  
Did not feel it at all — Felt it very strongly
2. I felt surrounded by the virtual world.  
Strongly disagree — Strongly agree
3. I felt like I was just perceiving pictures.  
Strongly disagree — Strongly agree
4. I did not feel present in the virtual space.  
Did not feel it at all — Felt it very strongly
5. I felt like I was acting within the virtual environment rather than operating it from the outside.  
Strongly disagree — Strongly agree
6. I felt present in the virtual space.  
Strongly disagree — Strongly agree
7. While exploring the virtual world, how aware were you of the real world surrounding you (e.g., ambient noise, room temperature, other people)?  
Very aware — Not aware at all
8. I was not aware of my real surroundings.  
Strongly disagree — Strongly agree
9. I was still aware of the real world while experiencing the virtual world.  
Strongly disagree — Strongly agree
10. I was completely immersed in the virtual world.  
Strongly disagree — Strongly agree
11. How real did the virtual world seem to you?  
Very real — Not real at all
12. How much did your experience in the virtual environment match your real-world experience?  
Did not match at all — Matched very well
13. How distinguishable was the virtual world from the real world?  
Completely virtual — Indistinguishable from the real world
14. The virtual world felt more realistic than the real world.  
Strongly disagree — Strongly agree